Data analytics for non life insurance (I)
GLM

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Contents

References

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Introduction

Introduction to GLM

- The objective of a priori ratemaking is to determine the way one or more key variables (number of claims, costs of claims,...) depend on a set of risk factors. That is to say, it consist on analysing the way a dependent variable Y varies with respect to a set of explanatory variables X in a regression model.

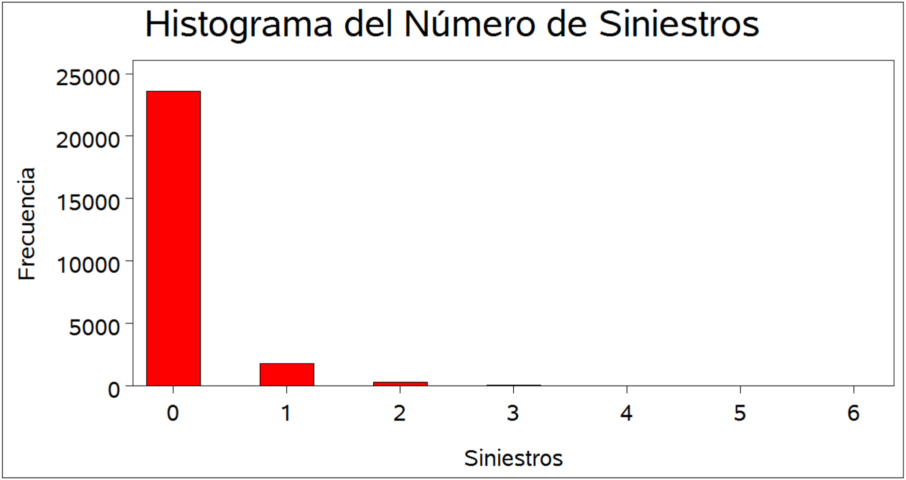
- We should take into account that the classical linear regression model is not suitable for *pricing* non life insurance products, because:

  (i) It assumes normality and homoscedasticity, while the number of claims has a discrete distribution and the cost of claims is a non-negative variable with positive skewness.

  (ii) In the linear model, the mean is a linear function of the explanatory variables while in the calculation of the premiums multiplicative models could be more suitable.

Introduction

Introduction to GLM



Introduction

Introduction to GLM

Generalized linear models (GLM) are an extention of the ordinary linear models in two directions:

a) **The probability distribution:** Instead of assuming a normal distribution for the error term in GLM we work with a general class of distributions (the exponential family) that includes a set of discrete and continuous distributions, such as the Normal, Binomial, Poisson, Gamma and Inverse Gaussian.

b) **Model for the mean** In the classical linear model the mean of the dependent variable is a linear function of the explanatory variables. In GLM a monotone transformation of the mean of the dependent variable is a linear function of the explanatory variables, with the linear and multiplicative relationship as particular cases.

Introduction

Components of a GLM

1. **Random component:** Let y_i , $i = 1, \ldots, n$, independent r.v. with density of the exponential family, that is to say:

$$f(y_i; \theta_i, \phi) = \exp\left\{ \frac{y_i\theta_i - b(\theta_i)}{\phi/\omega_i} + c(y, \phi) \right\}$$

in its natural or canonical form, where:

- ϕ is the dispersion parameter,
- ω_i is the weight ($\omega_i = 1$ if they are individual data, $\omega_i = n_i$ if they are averages, $\omega_i = 1/n_i$ if they are sums of n_i individual responses),
- θ_i is the canonical parameter,
- the functions $b(\cdot)$ y $c(\cdot)$ are known.

Introduction

Components of GLM

2. **Sistematic component:** The linear predictor is:

$$\eta_i = \sum_{j=1}^{k} X_{ij}\beta_j = \beta_1 + \beta_2 X_{i2} + \beta_3 X_{i3} + \cdots + \beta_k X_{ik}, \quad \forall i = 1, \ldots, n$$

   β the vector $k \times 1$ of parameters to estimate and X_{ij} the explanatory variables.

3. **Link component:** The link function g (*link*) that establishes the relationship of the expectation of the dependent variable with the linear predictor:

$$\eta_i = h^{-1}(\mu_i) = g(\mu_i), \ \forall i = 1, \ldots, n$$

   We have a **canonical link funcion** *(canonical link)* if $\theta_i = \eta_i$ $\forall i$.

Introduction

Moments

For the exponential family, the following relationships are fulfilled:

$$\mathrm{E}[\mathrm{y_i}] = \mathrm{b}'(\theta_\mathrm{i}) = \frac{\mathrm{db}(\theta)}{\mathrm{d}\theta} = \mu_\mathrm{i}$$

$$\mathrm{V}[\mathrm{y_i}] = \frac{\phi}{\omega_\mathrm{i}}\,\mathrm{b}''(\theta_\mathrm{i}) = \frac{\phi}{\omega_\mathrm{i}}\,\frac{\mathrm{d}^2\mathrm{b}(\theta)}{\mathrm{d}\theta^2} = \frac{\phi}{\omega_\mathrm{i}}\,\mathrm{V}(\mu)$$

where $\mathrm{V}(\cdot)$ is known as the **variance function**. This function establishes the relationship between the variance of Y and its mean μ.

Introduction

Family of models and links

Model	**Link**	$\boldsymbol{\eta}_\mathrm{i} = \mathrm{g}(\boldsymbol{\mu}_\mathrm{i})$	$\boldsymbol{\mu}_\mathrm{i} = \mathrm{g}^{-1}(\boldsymbol{\eta}_\mathrm{i})$	**Range of** $\mathrm{Y_i}$	$\mathrm{V}(\mathrm{Y_i}\mid\boldsymbol{\eta}_\mathrm{i})$
Normal	Identity	μ_i	η_i	$(-\infty, +\infty)$	ϕ
Binomial	Logit	$\ln \frac{\mu_\mathrm{i}}{1-\mu_\mathrm{i}}$	$\frac{\exp(\eta_\mathrm{i})}{1+\exp(\eta_\mathrm{i})}$	$\frac{0,1,\ldots,\mathrm{n_i}}{\mathrm{n_i}}$	$\frac{\mu_\mathrm{i}(1-\mu_\mathrm{i})}{\mathrm{n_i}}$
Binomial	Probit	$\Phi^{-1}(\mu_\mathrm{i})$	$\Phi(\eta_\mathrm{i})$	$\frac{0,1,\ldots,\mathrm{n_i}}{\mathrm{n_i}}$	$\frac{\mu_\mathrm{i}(1-\mu_\mathrm{i})}{\mathrm{n_i}}$
Poisson	Log	$\ln \mu_\mathrm{i}$	$\exp(\eta_\mathrm{i})$	$0, 1, 2, \ldots$	μ_i
Gamma	Inverse	${\mu_\mathrm{i}}^{-1}$	${\eta_\mathrm{i}}^{-1}$	$(0, \infty)$	$\phi\mu_\mathrm{i}^2$

The MLRM as a GLM

Definition

The MLRM:

$$y_i = \beta_1 x_{1i} + \beta_2 x_{2i} + \beta_3 x_{3i} + \ldots + \beta_k x_{ki} + \varepsilon_i \qquad \varepsilon_i \sim N(0, \sigma^2) \qquad i = 1 \ldots, n$$

The density funciton of y belongs to the exponential family:

$$f(y_i; \mu_i, \sigma) = \frac{1}{\sqrt{2\pi}\sigma} \exp\left\{-\frac{1}{2\sigma^2}(y_i - \mu_i)^2\right\} =$$

$$= \exp\left\{\frac{y_i\mu_i - \frac{\mu_i^2}{2}}{\sigma^2} - \frac{1}{2}\left[\ln(2\pi\sigma^2) + \frac{y_i^2}{\sigma^2}\right]\right\}$$

We have that $\theta_i = \mu_i$ and $\phi = \sigma^2$, and:

$$a(\phi) = \phi = \sigma^2, b(\theta_i) = \frac{\theta_i^2}{2} = \frac{\mu_i^2}{2}, c(y_i, \phi) = -\frac{1}{2}\left[\ln(2\pi\sigma^2) + \frac{y_i^2}{\sigma^2}\right]$$

and the link function (link) is the identity: $g(\mu_i) = \mu_i$.

The MLRM as a GLM

Moments

The moments are:

$$\mathrm{E[Y]} = \mathrm{b}'(\theta) = \frac{\mathrm{db}(\theta)}{\mathrm{d}\theta} = \frac{\mathrm{d}(\theta^2/2)}{\mathrm{d}\theta} = \theta = \mu$$

$$\mathrm{V[Y]} = \phi\,\mathrm{b}''(\theta) = \phi\,\frac{\mathrm{d}^2\mathrm{b}(\theta)}{\theta^2} = \phi\,\times 1 = \sigma^2$$

$$\mathrm{V}(\mu) = 1$$

Maximum Likelihood estimation of a GLM

Canonical link function (*canonical link*)

The log of the likelihood function for the individual observation y_i, taking into account the density of the exponential familty, is:

$$\ln \mathcal{L}(y_i; \theta_i, \phi) = \frac{y_i \theta_i - b(\theta_i)}{\phi/\omega_i} + c(y_i, \phi/\omega_i)$$

For n independient observations we have that:

$$\ln \mathcal{L}(y; \boldsymbol{\theta}, \phi) = \sum_{i=1}^{n} \left[\frac{y_i \theta_i - b(\theta_i)}{\phi/\omega_i} + c(y_i, \phi/\omega_i) \right]$$

Moreover, we have the link function: $g(\mu_i) = \eta_i = x_i' \boldsymbol{\beta}$, that, if it is canonical, we have that $\theta_i = \eta_i = x_i' \boldsymbol{\beta}$ and the log of the likelihood function is:

$$\ln \mathcal{L}(\boldsymbol{\beta}, \phi, \mid y) = \sum_{i=1}^{n} \left[\frac{y_i x_i' \boldsymbol{\beta} - b(x_i' \boldsymbol{\beta})}{\phi/\omega_i} + c(y_i, \phi/\omega_i) \right]$$

Maximum Likelihood estimation of a GLM

Canonical link function (*canonical link*)

Calculating the derivative with respect to $\boldsymbol{\beta}$:

$$\frac{\partial}{\partial \boldsymbol{\beta}} \ln \mathcal{L}(\boldsymbol{\beta}, \phi) = \frac{1}{\phi} \sum_{\mathrm{i}=1}^{\mathrm{n}} \left(\mathrm{y_i} - \mathrm{b}'(\mathrm{x}_{\boldsymbol{i}}' \boldsymbol{\beta})\right) \omega_{\mathrm{i}} \mathrm{x}_{\boldsymbol{i}}$$

As we have that $\mu_{\mathrm{i}} = \mathrm{b}'(\theta_{\mathrm{i}}) = \mathrm{b}'(\mathrm{x}_{\boldsymbol{i}}' \boldsymbol{\beta})$ we can get the ML estimates of the parameters $\hat{\boldsymbol{\beta}}_{\boldsymbol{MLE}}$ from the "*normal equations*":

$$\mathbf{0} = \sum_{\mathrm{i}=1}^{\mathrm{n}} \omega_{\mathrm{i}} \left(\mathrm{y_i} - \mu_{\mathrm{i}}\right) \mathrm{x}_{\boldsymbol{i}}$$

In general, this equations cannot be solved directly. The solution can be approximate by using iterative procedures such as the Newton-Raphson algorithm or the Fisher Scoring Method.

Maximum Likelihood estimation of a GLM

Variance calculation and inference

The variance-covariance matrix of the ML estimators can be obtained as the inverse of the information matrix, which is the expectation of the matrix of the negative second derivatives.

$$\widehat{\mathrm{Var}(\hat{\boldsymbol{\beta}})} = \left[\mathcal{I}(\hat{\boldsymbol{\beta}}\right]^{-1} = \frac{1}{\phi}\sum_{\mathrm{i}=1}^{\mathrm{n}} \omega_{\mathrm{i}} \mathrm{b}''(\mathrm{x}_{\boldsymbol{i}}'\hat{\boldsymbol{\beta}})\mathrm{x}_{\boldsymbol{i}}\mathrm{x}_{\boldsymbol{i}}'$$

The ML estimators are consistent and, asymptotically, have a normal distribution, therefore:

$$\frac{\hat{\boldsymbol{\beta}}}{\widehat{\mathrm{Var}(\hat{\boldsymbol{\beta}})}} \sim \mathrm{N}(0,1)$$

$$(\hat{\boldsymbol{\beta}} - \boldsymbol{\beta})' \left[\mathrm{Var}(\hat{\beta})\right]^{-1} (\hat{\boldsymbol{\beta}} - \boldsymbol{\beta}) \sim \chi^2_{\mathrm{k}-1}$$

Maximum Likelihood estimation of a GLM

Estimating the dispersion parameter

It is not necessary to estimate the dispersion parameter ϕ to obtain the ML estimations of the parameters $\boldsymbol{\beta}$. Moreover, el the dispersion parameter ϕ is known for some distributions, such as the Binomial or Poisson, and we do not need to estimate it.

The next expression provides with a consistent estimator of ϕ by using the ML estimators of β:

$$\hat{\phi} = \frac{1}{\mathrm{n}-\mathrm{k}} \sum_{\mathrm{i}=1}^{\mathrm{n}} \frac{\omega_{\mathrm{i}}(\mathrm{y}_{\mathrm{i}} - \mathrm{b}'(\mathrm{x}_{\boldsymbol{i}}'\hat{\boldsymbol{\beta}}))^2}{\mathrm{b}''(\mathrm{x}_{\boldsymbol{i}}'\hat{\boldsymbol{\beta}})}$$

Goodness of fit of a GLM

Deviance

- The Deviance (**Deviance**) is a goodness of fit measure of a GLM. Lower values of the deviance indicate a better fit.
- The deviance compares the fit of a model with the fit of the **saturated model**.
- The saturated model has the same shape as the fitted model, but it has as many parameters as observations, $\theta_i, \; i = 1, \ldots, n$, which represents the best possible fit.
- By taking the first derivatives of the logarithm of the likelihood function with respect to θ_i and making them to be equal to 0:

$$\frac{\partial}{\partial \theta_i} \ln \mathcal{L}(y_i; \theta_i \phi) = \frac{y_i - b'(\theta_i)}{\phi} = 0$$

we obtain the estimation of $\theta_{i,SAT}$ as the solution for $y_i = b'(\theta_{i,SAT})$.

Goodness of fit of a GLM

Deviance

- Then, the likelihood of the saturated model $\mathcal{L}(\boldsymbol{\theta}_{\boldsymbol{SAT}})$ will be the highest possible value of the logarithm of the likelihood.
- The **scaled deviance** (*scaled deviance*) can be obtained by using the likelihood ratio test.

$$\mathrm{D}^*(\hat{\boldsymbol{\theta}}) = -2\left(\ln \mathcal{L}(\hat{\boldsymbol{\theta}}) - \ln \mathcal{L}(\boldsymbol{\theta}_{\boldsymbol{SAT}})\right) \sim \chi^2_{\mathrm{n-k}}$$

- The **Non scaled deviance** (*deviance*) we should multiply by the scale factor ϕ:

$$\mathrm{D}(\hat{\boldsymbol{\theta}}) = \phi\, \mathrm{D}^*(\hat{\boldsymbol{\theta}})$$

Goodness of fit of a GLM

Deviance

- The expression of the deviance for some specific models:
  - Normal: $\mathrm{D}(\hat{\boldsymbol{\mu}}) = \sum(\mathrm{y_i} - \hat{\mu}_\mathrm{i})^2$
  - Binomial: $\mathrm{D}(\hat{\boldsymbol{\mu}}) = 2\sum\left\{\mathrm{y_i}\ln\frac{\mathrm{y_i}}{\hat{\mu}_\mathrm{i}} + (\mathrm{n_i} - \mathrm{y_i})\ln\frac{\mathrm{n_i}-\mathrm{y_i}}{\mathrm{n_i}-\hat{\mu}_\mathrm{i}}\right\}$ with $\hat{\mu}_\mathrm{i} = \mathrm{n_i}\hat{\mathrm{p}}_\mathrm{i}$
  - Poisson: $\mathrm{D}(\hat{\boldsymbol{\mu}}) = 2\sum\left\{\mathrm{y_i}\ln\frac{\mathrm{y_i}}{\hat{\mu}_\mathrm{i}} - (\mathrm{y_i} - \hat{\mu}_\mathrm{i})\right\}$ with and $\ln \mathrm{y} = 0$ si $\mathrm{y} = 0$
- It is also possible to use the AIC (Akaike Information Criterion) and the BIC (Bayesian Information Criterion), that take into account the number of parameters included in the model. Models with lower values of AIC and BIc are better. The expressions are:
  - $AIC = -2\ln\mathcal{L} + 2p$
  - $BIC = -2\ln\mathcal{L} + p\ln n$

Count data models: Poisson

Introduction

- The *count data* models are those with a dependent variable, y, counts the number of times that an event occurs, and therefore, it takes discrete values: $0, 1, 2, \ldots$.
- Count data models are used in acturial science for the modelization of the number of claims.
- The Poisson model, which is a GLM, is suitable for many data sets frequenly used by actuaries.
- Nevertheless, the Poisson distribution has mean equal to variance, and it can be an important restriction. Alternative, and more general, modelizations are possible, such as the Negative Binomial.

Count data models: Poisson

Especification

- The probability law of a Poisson distribution is given by:

$$\mathrm{Pr}(\mathrm{y}=\mathrm{j}) = \frac{\mu^{\mathrm{j}}}{\mathrm{j}!}\exp(-\mu)\,, \quad \mathrm{j}=0,1,2,\ldots$$

  μ is the number of times that an event occurs in a **time interval**.

- $\mathrm{E}(\mathrm{y}) = \mu$ is the average of the distribution, and also the variance $\mathrm{Var}(\mathrm{y}) = \mu$.
- We want that the mean μ could vary depending on the values of some explanatory variables thar represent risk factors, which is specified as: $\mathrm{E}(\mathrm{y_i}) = \mu_\mathrm{i} = \exp(\mathrm{x}_\mathrm{i}'\boldsymbol{\beta})$.
- Moreover, the mean also varies according to the **risk exposition**, $\mathrm{t_i}$, that is to say, the fraction of the year that the insured is covered by the policy, such that $\mathrm{E}(\mathrm{y_i}) = \mu_\mathrm{i} = \mathrm{t_i}\,\exp(\mathrm{x}_\mathrm{i}'\boldsymbol{\beta})$.
- Using the risk exposition is justified by the fact that the expected number of claims is directly proportional to the coverage period of the policy.

Count data models: Poisson

Specification as a GLM

Let $\mathrm{Y_i} \sim \mathcal{P}\text{oisson}(\mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta}))\ \ \mathrm{i} = 1, \ldots, \mathrm{n}$, i.i.d., then:

$$\mathrm{Pr}(\mathrm{Y_i} = \mathrm{y_i}) = \frac{(\mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta}))^{\mathrm{y_i}}}{\mathrm{y_i}!} \exp(-\mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta}))$$

that can also be expressed as:

$$\mathrm{Pr}(\mathrm{Y_i} = \mathrm{y_i}) = \exp\left\{\mathrm{y_i} \ln(\mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta})) - \ln \mathrm{y_i}! - \mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta})\right\}$$

by ordering the terms, we have that:

$$\mathrm{Pr}(\mathrm{Y_i} = \mathrm{y_i}) = \exp\left\{\mathrm{y_i} \ln(\mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta})) - \mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta}) - \ln \mathrm{y_i}!\right\}$$

threfore, it belongs to the exponential family, as:

$\theta_\mathrm{i} = \ln(\mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta}))$ $\quad \mathrm{b}(\theta_\mathrm{i}) = -\mathrm{t_i}\ \exp(\mathrm{x_i'}\boldsymbol{\beta}) = \exp(\theta_\mathrm{i})$ $\quad \mathrm{c}(\mathrm{y};\phi) = -\ln \mathrm{y_i}!$ $\quad \phi = 1$

and the link function (link) is: $\mathrm{g}(\mu_\mathrm{i}) = \ln(\mu_\mathrm{i}) = \ln \mathrm{t_i} + \mathrm{x_i'}\boldsymbol{\beta}$.

Count data models: Poisson

ML estimatio of a Poisson Model

The log likelihood function is:

$$\ln \mathcal{L}(\boldsymbol{\beta} \mid \mathrm{y}) = \sum_{\mathrm{i}=1}^{\mathrm{n}} \left[-\mathrm{t_i} \exp(\mathrm{x_i'}\boldsymbol{\beta}) + \mathrm{y_i}(\ln \mathrm{t_i} + \mathrm{x_i'}\boldsymbol{\beta}) - \ln \mathrm{y_i}!\right]$$

if the first derivatives wrt. the parameters are zero, then we have that:

$$\frac{\partial}{\partial \boldsymbol{\beta}} \ln \mathcal{L}(\boldsymbol{\beta}) = \sum_{\mathrm{i}=1}^{\mathrm{n}} (\mathrm{y_i} - \mathrm{t_i} \exp(\mathrm{x_i'}\hat{\boldsymbol{\beta}}))\mathrm{x_i} = 0 \quad \text{where } \hat{\boldsymbol{\beta}} \text{ is the ML estimator.}$$

If we calculate the second derivatives, we obtain the Information matrix, that makes it possible to have an estimation of the covariance matrix of $\hat{\boldsymbol{\beta}}$:

$$\mathcal{I}(\boldsymbol{\beta}) = -\mathrm{E}\frac{\partial^2}{\partial \boldsymbol{\beta} \partial \boldsymbol{\beta}'} \ln \mathcal{L}(\boldsymbol{\beta}) = \sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{t_i} \exp(\mathrm{x_i'}\hat{\boldsymbol{\beta}})\mathrm{x_i}\mathrm{x_i'}$$

$$\widehat{\mathrm{Var}(\hat{\boldsymbol{\beta}})} = \left(\sum_{\mathrm{i}=1}^{\mathrm{n}} \mathrm{t_i} \exp(\mathrm{x_i'}\hat{\boldsymbol{\beta}})\mathrm{x_i}\mathrm{x_i'} \right)^{-1}$$

Count data models: Poisson

Validation of a Poisson model

In GLM we use the ***Pearson residuals***, $\mathrm{r_i}$, instead of the classical residuals:

$$\mathrm{r_i} = \frac{\mathrm{y_i} - \hat{\mu}_\mathrm{i}}{\sqrt{\hat{\mu}_\mathrm{i}}} = \frac{\mathrm{y_i} - \mathrm{t_i}\ \exp(\mathrm{x_i'}\hat{\boldsymbol{\beta}})}{\sqrt{\mathrm{t_i}\ \exp(\mathrm{x_i'}\hat{\boldsymbol{\beta}})}}$$

These residuals let us build the **Pearson statistic**, that, under the null hypothesis of a good fit, is χ^2 distributed:

$$\sum_{\mathrm{i=1}}^{\mathrm{n}} \mathrm{r_i^2} = \sum_{\mathrm{i=1}}^{\mathrm{n}} \frac{(\mathrm{y_i} - \hat{\mu}_\mathrm{i})^2}{\hat{\mu}_\mathrm{i}} = \frac{(\mathrm{y_i} - \mathrm{t_i}\ \exp(\mathrm{x_i'}\hat{\boldsymbol{\beta}}))^2}{\mathrm{t_i}\ \exp(\mathrm{x_i'}\hat{\boldsymbol{\beta}})} \sim \chi^2_{\mathrm{n-k}}$$

Another possibility in GLM is to calculate the ***Deviance residuals***, $\mathrm{dr_i}$:

$$\mathrm{dr_i} = \mathrm{signo}(\mathrm{y_i} - \hat{\mu}_\mathrm{i})\sqrt{2(\ln \mathcal{L}(\hat{\boldsymbol{\beta}}_{\boldsymbol{SAT}}) - \ln \mathcal{L}(\hat{\boldsymbol{\beta}}))}$$

Count data models: Poisson

Validation of a Poisson model

On the other hand, we can also use the Deviance, that, in the case of a Poisson model, has the following expression:

$$\mathrm{D}(\hat{\boldsymbol{\mu}}) = 2 \sum \left\{ \mathrm{y_i} \ln \left(\frac{\mathrm{y_i}}{\hat{\mu}_\mathrm{i}} \right) - (\mathrm{y_i} - \hat{\mu}_\mathrm{i}) \right\} \sim \chi^2_{\mathrm{n-k}}$$

where $\hat{\mu}_\mathrm{i} = \mathrm{t_i} \exp(\mathrm{x}_\mathrm{i}' \hat{\boldsymbol{\beta}})$. In general, the higher the deviance, the worse the fit.

Moreover, we can also use the Likelihood Ratio Test to compare nested models:

$$\mathrm{RV} = -2 \left(\ln \mathcal{L}(\hat{\boldsymbol{\beta}}_\mathrm{R}) - \ln \mathcal{L}(\hat{\boldsymbol{\beta}}) \right) \sim \chi^2_\mathrm{r}$$

where $\ln \mathcal{L}(\hat{\boldsymbol{\beta}}_\mathrm{R})$ is the logarithm of the likelihood of the estimated model with r linear constrains on the parameters.

Count data models: Poisson

Poisson model: Example

Example: Number of motor claims (n=7483)
(source: General Insurance Association of Singapore, 1993)

Variables:

- **Clm_count**: Number of claims during the year 1993
- **Female**: Gender of the insured (1 woman, 0 man)
- **Auto**: Vehicle type (1 automobile, 0 other)
- **Vagecat1**: Vehicle age in years (seven cathegories 0, 1, 2, 3-5, 6-10, 11-15, 16 or more)
- **Agecat**: Age of the policyholder (seven cathegories: less than 21, 22-25, 26-35, 36-45, 46-55, 56-65, 66 or more)
- **Ncd**: No claim discounts (six cathegories: 0, 10, 20, 30, 40, 50)
- **Lnweight**: Logarithm of the risk exposition (fraction of the year with policy coverage)

Count data models: Poisson

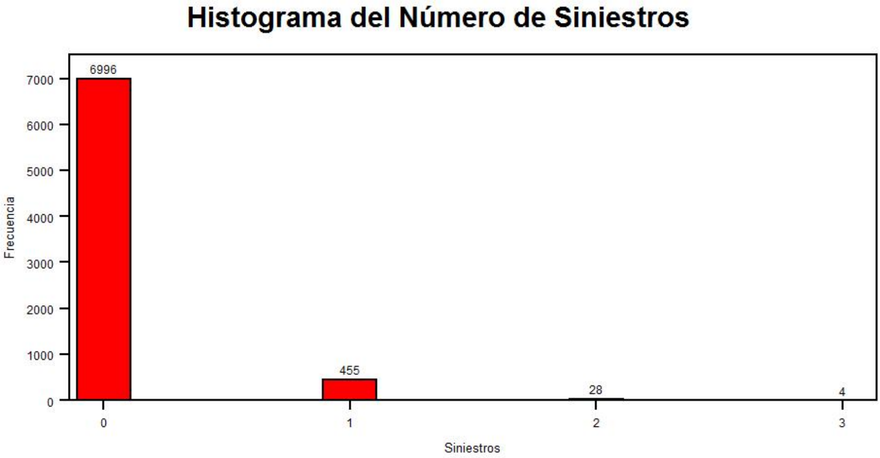
Poisson model: Example

By using **PROC GENMOD** in SAS:

```
/* HISTOGRAM of Clm_count */
proc univariate data=Singapore noprint;
 Histogram Clm_count / vscale=count midpoints=0 to 3 by 1
                      vaxislabel='Frecuencia' cfill=red
                      barwidth=6 barlabel=count;
 title 'Histograma del Numero de Siniestros';
run;
quit;
/* POISSON MODEL ESTIMATION*/
proc genmod data=singapore;
class ncd agecat vagecat1;
model clm_count=female auto vagecat1 auto*agecat auto*ncd
                  / dist=poisson link=log offset=lnweight;
output out=resid pred=pred;
run;
```

Count data models: Poisson

Poisson model: Example



Count data models: Poisson

Poisson model: Example

```
                    Procedimiento GENMOD
                   Información del modelo

          Conjunto de datos       WORK.SINGAPORE
          Distribución                   Poisson
          Función de vínculo                 Log
          Variable dependiente         Clm_Count
          Variable Offset               LNWEIGHT

        Número de observaciones leídas        7483
        Número de observaciones usadas        7483

             Información de nivel de clase
            Clase        Niveles    Valores
            NCD                6    0 10 20 30 40 50
            AgeCat             7    0 2 3 4 5 6 7
            VAgecat1           5    2 3 4 5 6

        Criterios para valorar la bondad de ajuste
Criterio                       DF         Valor       Valor/DF
Desviación                   7466     2611.4599         0.3498
Desviación escalada          7466     2611.4599         0.3498
Chi-cuadrado de Pearson      7466     7441.7879         0.9968
Pearson X2 escalado          7466     7441.7879         0.9968
Verosimilitud log                    -1776.7304
Verosimilitud log completa           -1803.3055
AIC (mejor más pequeño)               3640.6110
AICC (mejor más pequeño)              3640.6930
BIC (mejor más pequeño)               3758.2577

Algoritmo convergido.
```

Count data models: Poisson

Poisson model: Example

Análisis de estimadores de parámetros de máxima verosimilitud

Parámetro		DF	Estimador	Error estándar	Wald 95% Límites de confianza		Chi-cuadrado de Wald	Pr > ChiSq
Intercept		1	-3.3061	0.5008	-4.2877	-2.3246	43.58	<.0001
Female		1	-0.1730	0.1551	-0.4770	0.1311	1.24	0.2648
auto		1	0.0794	0.7213	-1.3343	1.4932	0.01	0.9123
VAgecat1	2	1	1.6739	0.5110	0.6724	2.6754	10.73	0.0011
VAgecat1	3	1	1.5038	0.5155	0.4935	2.5141	8.51	0.0035
VAgecat1	4	1	1.0807	0.5186	0.0642	2.0971	4.34	0.0372
VAgecat1	5	1	0.3619	0.5311	-0.6790	1.4028	0.46	0.4956
VAgecat1	6	0	0.0000	0.0000	0.0000	0.0000	.	.
auto*AgeCat	0	0	0.0000	0.0000	0.0000	0.0000	.	.
auto*AgeCat	2	1	-0.7466	0.7765	-2.2686	0.7754	0.92	0.3364
auto*AgeCat	3	1	-0.5815	0.7180	-1.9888	0.8258	0.66	0.4180
auto*AgeCat	4	1	-0.6230	0.7160	-2.0264	0.7804	0.76	0.3843
auto*AgeCat	5	1	-0.8033	0.7308	-2.2355	0.6290	1.21	0.2717
auto*AgeCat	6	1	-0.2574	0.7499	-1.7273	1.2124	0.12	0.7314
auto*AgeCat	7	0	0.0000	0.0000	0.0000	0.0000	.	.
auto*NCD	0	1	0.7293	0.1550	0.4254	1.0331	22.13	<.0001
auto*NCD	10	1	0.5280	0.1932	0.1493	0.9068	7.47	0.0063
auto*NCD	20	1	0.2929	0.2209	-0.1400	0.7257	1.76	0.1849
auto*NCD	30	1	0.2597	0.2254	-0.1822	0.7015	1.33	0.2494
auto*NCD	40	1	-0.0945	0.2763	-0.6361	0.4471	0.12	0.7323
auto*NCD	50	0	0.0000	0.0000	0.0000	0.0000	.	.
Escal		0	1.0000	0.0000	1.0000	1.0000		

NOTE: El parámetro de escala se ha mantenido fijo.

Count data models: Poisson

Poisson model: Example

By using R:

```
>#  ESTIMATING THE POISSON MODEL
>Auto = 1*(VehicleType=="A")
>Female = 1*(SexInsured == "F" )
>NCD1F = relevel(factor(NCD), ref="50")
>AgeCatF = relevel(factor(AgeCat), ref="7")
>VAgecat1F = relevel(factor(VAgecat1), ref="6")
>CountPoisson = glm(Clm_Count~Female+Auto
+                    +Auto:AgeCatF + Auto:NCD1F + VAgecat1F,
+                   offset=LNWEIGHT,poisson(link=log))
>summary(CountPoisson)
```

Count data models: Poisson

Poisson model: Example

```
Call:
glm(formula = Clm_Count ~ Female + Auto + Auto:AgeCatF + Auto:NCD1F +
    VAgecat1F, family = poisson(link = log), offset = LNWEIGHT)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-0.82344  -0.43526  -0.31982  -0.21001   3.69627

Coefficients: (1 not defined because of singularities)
               Estimate Std. Error z value  Pr(>|z|)
(Intercept)   -3.306145   0.500765 -6.6022 4.051e-11 ***
Female        -0.172962   0.155122 -1.1150  0.264849
Auto           0.079433   0.721326  0.1101  0.912313
VAgecat1F2     1.673892   0.510954  3.2760  0.001053 **
VAgecat1F3     1.503765   0.515452  2.9174  0.003530 **
VAgecat1F4     1.080656   0.518601  2.0838  0.037179 *
VAgecat1F5     0.361927   0.531051  0.6815  0.495537
Auto:AgeCatF0        NA         NA      NA        NA
Auto:AgeCatF2 -0.746568   0.776542 -0.9614  0.336351
Auto:AgeCatF3 -0.581534   0.718022 -0.8099  0.417991
Auto:AgeCatF4 -0.623014   0.716047 -0.8701  0.384260
Auto:AgeCatF5 -0.803269   0.730768 -1.0992  0.271676
Auto:AgeCatF6 -0.257442   0.749945 -0.3433  0.731387
Auto:NCD1F0    0.729274   0.155030  4.7041 2.550e-06 ***
Auto:NCD1F10   0.528037   0.193246  2.7325  0.006286 **
Auto:NCD1F20   0.292857   0.220866  1.3259  0.184857
Auto:NCD1F30   0.259660   0.225442  1.1518  0.249411
Auto:NCD1F40  -0.094515   0.276345 -0.3420  0.732336
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for poisson family taken to be 1)
```

```
    Null deviance: 2716.87  on 7482  degrees of freedom
Residual deviance: 2611.46  on 7466  degrees of freedom
AIC: 3640.61

Number of Fisher Scoring iterations: 6

Table1p
[1,] 6986.9426 470.29514 24.627175 1.0913168 0.042243132

PersonG
8.76585
```

Count data models: Negative Binomial

Negative Binomial as a mixture of Poisson and Gamma

- Let y be a count r.v. Poisson distributed with mean $\boldsymbol{\lambda}$.

$$\mathrm{y} \mid \lambda \sim \mathcal{P}\mathrm{oisson}(\lambda) \Longrightarrow \mathrm{f}(\mathrm{y} \mid \lambda) = \frac{\mathrm{e}^{-\lambda}\lambda^{\mathrm{y}}}{\mathrm{y}!}$$

- We assume that $\boldsymbol{\lambda}$ is not constant but varies among individuals.

- Therefore $\boldsymbol{\lambda}$ is a r.v. and we assume that is Gamma distributed with parameters a and b

$$\lambda \sim \mathcal{G}\mathrm{amma}(\mathrm{a},\mathrm{b}) \Longrightarrow \mathrm{g}(\lambda) = \frac{\mathrm{b}^{\mathrm{a}}\mathrm{e}^{-\mathrm{b}\lambda}\lambda^{\mathrm{a}-1}}{\Gamma(\mathrm{a})}$$

with $\mathrm{E}(\lambda) = \dfrac{\mathrm{a}}{\mathrm{b}}$ and $\mathrm{Var}(\lambda) = \dfrac{\mathrm{a}}{\mathrm{b}^2}$.

Count data models: Negative Binomial

Negative Binomial as a mixture of Poisson and Gamma

We obtain the **Negative Binomial** as a mixture of Poisson and Gamma:

$$f(y \mid a,b) = \int_0^\infty f(y \mid \lambda) g(\lambda \mid a,b) d\lambda =$$

$$= \int_0^\infty \frac{e^{-\lambda}(\lambda)^y}{y!} \frac{b^a e^{-b\lambda} \lambda^{a-1}}{\Gamma(a)} d\lambda = \frac{b^a}{y!\Gamma(a)} \int_0^\infty e^{-(1+b)\lambda} \lambda^{y+a-1} d\lambda =$$

$$= \frac{b^a}{y!\Gamma(a)(1+b)^{y+a}} \int_0^\infty e^{-(1+b)\lambda} [(1+b)\lambda]^{y+a-1} d[(1+b)\lambda]$$

[We know that $\Gamma(a) = \int_0^\infty t^{a-1} e^{-t} dt$ y $\Gamma(k+1) = k\Gamma(k) = k!$ if k integer]

$$f(y \mid a,b) = \frac{\Gamma(y+a)}{\Gamma(y+1)\Gamma(a)} \left(\frac{b}{1+b}\right)^a \left(\frac{1}{1+b}\right)^y$$

with $E(y) = \frac{a}{b} = \bar{\lambda}$ and $Var(y) = \frac{a}{b}\left(1 + \frac{1}{b}\right) = \bar{\lambda}\left(1 + \frac{\bar{\lambda}}{a}\right)$.

Count data models: Negative Binomial

Specification of the Negative Binomial model

- As we did for the Poisson model, we aim that the expected number of claims for a time unit, μ_{i}, may vary depending on the values of some explanatory variables that represent different risk factors.
- Nevertheless, we now introduce an heterogeneity factor, ε_{i}, as this heterogeneity among individuals is not appropriately captured by the explanatory variables. We specify it in the following way:

$$\mathrm{E}(\mathrm{y_i}) = \mu_{\mathrm{i}} = \exp(\mathrm{x_i}'\boldsymbol{\beta} + \boldsymbol{\varepsilon}_{\mathrm{i}})$$

  where $\mathrm{u_i} = \exp(\varepsilon_{\mathrm{i}})$ is a r.v., Gamma distributed $\mathrm{E}(\mathrm{u_i}) = 1$ and $\mathrm{Var}(\mathrm{u_i}) = 1/\mathrm{a}$, that is to say, $\mathrm{u_i} \sim \mathcal{G}\mathrm{amma}(\mathrm{a}, \mathrm{a})$.
- Then, the specification of the probability will be:

$$\mathrm{f}(\mathrm{y} \mid \mathrm{a}) = \frac{\Gamma(\mathrm{y}+\mathrm{a})}{\Gamma(\mathrm{y}+1)\Gamma(\mathrm{a})} \left(\frac{\exp(\mathrm{x_i}'\boldsymbol{\beta})}{\mathrm{a}}\right)^{\mathrm{y}} \left(1 + \frac{\exp(\mathrm{x_i}'\boldsymbol{\beta})}{\mathrm{a}}\right)^{-(\mathrm{y}+\mathrm{a})}$$

Count data models: Negative Binomial

ML estimation

The log of the likelihood function is:

$$\ln \mathcal{L}(\boldsymbol{\beta}, \mathrm{a}) = \sum_{i=1}^{n} \left\{ \left(\sum_{j=0}^{y-1} \ln(j+a) \right) - \ln y_i! - (y_i + a) \ln \left(1 + \frac{e^{x_i' \boldsymbol{\beta}}}{a} \right) + y_i \ln \left(\frac{e^{x_i' \boldsymbol{\beta}}}{a} \right) \right\}$$

The ML estimators of the parameters $\boldsymbol{\beta}$ and a are obtained by solving:

$$\frac{\partial}{\partial \boldsymbol{\beta}} \ln \mathcal{L}(\boldsymbol{\beta}, a) = \sum_{i=1}^{n} \frac{y_i - \exp(x_i' \hat{\boldsymbol{\beta}})}{1 + \frac{\exp(x_i' \hat{\boldsymbol{\beta}})}{\hat{a}}} x_i = \mathbf{0}$$

$$\frac{\partial}{\partial a} \ln \mathcal{L}(\boldsymbol{\beta}, a) = \sum_{i=1}^{n} \left\{ \hat{a}^2 \left[\ln \left(1 + \frac{\exp x_i' \hat{\boldsymbol{\beta}}}{\hat{a}} \right) - \sum_{j=0}^{y-1} \frac{1}{j + \hat{a}} \right] + \frac{\hat{a}(y_i - \exp(x_i' \hat{\boldsymbol{\beta}}))}{1 + \frac{\exp(x_i' \hat{\boldsymbol{\beta}})}{\hat{a}}} \right\} = 0$$

where $\hat{\boldsymbol{\beta}}$ and $\hat{a}$ are the ML estimators.

Count data models: Negative Binomial

Example

We can use the **PROC GENMOD** of SAS:

```
/* Estimating the NB model */
proc genmod data=singapore;
class ncd agecat vagecat1;
model clm_count=female auto auto*agecat vagecat1 auto*ncd
               /dist=nb link=log offset=lnweight;
output out=resid4 pred=pred;
run;
```

Count data models: Negative Binomial

Example

The GENMOD Procedure

Model Information

Data Set	WORK.SINGAPORE
Distribution	Negative Binomial
Link Function	Log
Dependent Variable	Clm_Count
Offset Variable	LNWEIGHT

Number of Observations Read	7483
Number of Observations Used	7483

Class Level Information

Class	Levels	Values
ncd	6	0 10 20 30 40 50
agecat	7	0 2 3 4 5 6 7
vagecat1	5	2 3 4 5 6

Criteria For Assessing Goodness Of Fit

Criterion	DF	Value	Value/DF
Deviance	7466	2413.2243	0.3232
Scaled Deviance	7466	2413.2243	0.3232
Pearson Chi-Square	7466	7215.5253	0.9665
Scaled Pearson X2	7466	7215.5253	0.9665
Log Likelihood		-1774.4939	
Full Log Likelihood		-1801.0691	
AIC (smaller is better)		3638.1382	
AICC (smaller is better)		3638.2298	
BIC (smaller is better)		3762.7052	

Algorithm converged.

Count data models: Negative Binomial

Example

Analysis Of Maximum Likelihood Parameter Estimates

Parameter		DF	Estimate	Standard Error	Wald 95% Confidence Limits		Wald Chi-Square	Pr > ChiSq
Intercept		1	-3.3081	0.5034	-4.2948	-2.3213	43.18	<.0001
Female		1	-0.1764	0.1587	-0.4873	0.1346	1.24	0.2663
auto		1	0.0822	0.7453	-1.3785	1.5428	0.01	0.9122
auto*AgeCat	0	0	0.0000	0.0000	0.0000	0.0000	.	.
auto*AgeCat	2	1	-0.7420	0.8019	-2.3136	0.8296	0.86	0.3548
auto*AgeCat	3	1	-0.5852	0.7426	-2.0406	0.8702	0.62	0.4307
auto*AgeCat	4	1	-0.6243	0.7403	-2.0754	0.8267	0.71	0.3991
auto*AgeCat	5	1	-0.8055	0.7551	-2.2855	0.6745	1.14	0.2861
auto*AgeCat	6	1	-0.2369	0.7754	-1.7566	1.2828	0.09	0.7600
auto*AgeCat	7	0	0.0000	0.0000	0.0000	0.0000	.	.
VAgecat1	2	1	1.6732	0.5140	0.6657	2.6807	10.60	0.0011
VAgecat1	3	1	1.5029	0.5187	0.4862	2.5196	8.39	0.0038
VAgecat1	4	1	1.0809	0.5217	0.0583	2.1035	4.29	0.0383
VAgecat1	5	1	0.3627	0.5340	-0.6840	1.4093	0.46	0.4970
VAgecat1	6	0	0.0000	0.0000	0.0000	0.0000	.	.
auto*NCD	0	1	0.7336	0.1589	0.4222	1.0450	21.33	<.0001
auto*NCD	10	1	0.5315	0.1978	0.1438	0.9192	7.22	0.0072
auto*NCD	20	1	0.2973	0.2255	-0.1446	0.7393	1.74	0.1873
auto*NCD	30	1	0.2684	0.2301	-0.1825	0.7193	1.36	0.2433
auto*NCD	40	1	-0.0882	0.2805	-0.6380	0.4616	0.10	0.7532
auto*NCD	50	0	0.0000	0.0000	0.0000	0.0000	.	.
Dispersion		1	0.4289	0.2351	0.1465	1.2557		

NOTE: The negative binomial dispersion parameter was estimated by maximum likelihood.

Count data models: Negative Binomial

Example

By using R:

```
># instalation of the "MASS" package
>install.packages("MASS")
>library(MASS)

>#  Estimating the Negative Binomial model;
>CountNB1 = glm.nb(Clm_Count~Female+Auto
+                  +Auto:AgeCatF + Auto:NCD1F
+                  +VAgecat1F+offset(LNWEIGHT),link=log)
>summary(CountNB1)
```

Count data models: Negative Binomial

Example

```
Call:
glm.nb(formula = Clm_Count ~ Female + Auto + Auto:AgeCatF + Auto:NCD1F +
    VAgecat1F + offset(LNWEIGHT), link = log, init.theta = 2.331455414)

Deviance Residuals:
     Min        1Q    Median        3Q       Max
-0.80500  -0.43052  -0.31797  -0.20939   3.33869

Coefficients: (1 not defined because of singularities)
               Estimate Std. Error z value  Pr(>|z|)
(Intercept)   -3.308061   0.503575 -6.5692 5.060e-11 ***
Female        -0.176355   0.158584 -1.1121  0.266114
Auto           0.082159   0.745232  0.1102  0.912214
VAgecat1F2     1.673218   0.514274  3.2536  0.001140 **
VAgecat1F3     1.502928   0.518908  2.8963  0.003776 **
VAgecat1F4     1.080923   0.521902  2.0711  0.038347 *
VAgecat1F5     0.362673   0.534182  0.6789  0.497181
Auto:AgeCatF0        NA         NA      NA        NA
Auto:AgeCatF2 -0.741988   0.800622 -0.9268  0.354049
Auto:AgeCatF3 -0.585169   0.741846 -0.7888  0.430228
Auto:AgeCatF4 -0.624333   0.739805 -0.8439  0.398716
Auto:AgeCatF5 -0.805530   0.754570 -1.0675  0.285731
Auto:AgeCatF6 -0.236913   0.774064 -0.3061  0.759556
Auto:NCD1F0    0.733596   0.158559  4.6266 3.716e-06 ***
Auto:NCD1F10   0.531504   0.197686  2.6886  0.007175 **
Auto:NCD1F20   0.297313   0.225193  1.3203  0.186749
Auto:NCD1F30   0.268431   0.229521  1.1695  0.242191
Auto:NCD1F40  -0.088190   0.279975 -0.3150  0.752766
---
Signif. codes:  0 ‘***’ 0.001 ‘**’ 0.01 ‘*’ 0.05 ‘.’ 0.1 ‘ ’ 1

(Dispersion parameter for Negative Binomial(2.3315) family taken to be 1)
```

```
    Null deviance: 2515.28  on 7482  degrees of freedom
Residual deviance: 2413.22  on 7466  degrees of freedom
AIC: 3638.14

Number of Fisher Scoring iterations: 1


              Theta:  2.33
          Std. Err.:  1.28

 2 x log-likelihood:  -3602.138

Table1p
[1,] 6997.0096 451.64314 31.715237 2.417318 0.19624484

PersonG
1.6927806
```

Zero inflated models

Introduction

- Very frequently we find "excess" of zeroes. This happens because many insureds do not have any claim or sometimes they prefer not to declare them (if they are small) to avoid future premium increases.

- The ZI model represents the probability of the number of claims $\mathrm{Pr}(\mathrm{y_i}=\mathrm{j})$ as a mixture between the probability that the i-th observation is equal to zero and the probability that the t-th observation comes from a Poisson or Negative Binomial distribution. That is to say, its probability distribution is given by:

$$\mathrm{Pr}(\mathrm{y_i}=\mathrm{j}) = \begin{cases} \pi_\mathrm{i} + (1-\pi_\mathrm{i})\mathrm{g_i}(0) & \mathrm{j}=0 \\ \\ (1-\pi_\mathrm{i})\mathrm{g_i}(\mathrm{j}) & \mathrm{j}=1,2,\ldots \end{cases}$$

Zero inflated models

Zero Inflated Poisson Regression Model (ZIP)

- The ZIP regression models is specified as follows:

$$g(y_i \mid \lambda_i) = \frac{e^{-\lambda_i}\lambda_i^{y_i}}{y_i!} \qquad \lambda_i = \exp(x_i'\boldsymbol{\beta}) \qquad \pi_i = \frac{\exp(z_i'\boldsymbol{\gamma})}{1+\exp(z_i'\boldsymbol{\gamma})}$$

$$\Pr(y_i) = \begin{cases} \pi_i + (1-\pi_i)e^{-\lambda_i} & y_i = 0 \\ (1-\pi_i)\frac{e^{-\lambda_i}\lambda_i^{y_i}}{y_i!} & y_i = 1, 2, \ldots \end{cases}$$

with:

$$E(y_i) = \mu_i = (1-\pi_i)\lambda_i$$

$$\mathrm{Var}(y_i) = \mu_i + \frac{\pi_i}{1-\pi_i}\mu_i^2 = (1-\pi_i)\lambda_i + \pi_i(1-\pi_i)\lambda_i^2$$

Zero inflated models

Zero Inflated Negative Binomial Regression Model (ZINB)

- The Zero Inflated Negative Binomial regression model (ZINB) is specified as:

$$g(y_i \mid \lambda_i) = \frac{\Gamma(a+y)}{\Gamma(y+1)\Gamma(a)} \left(\frac{a}{a+\lambda_i}\right)^a \left(\frac{\lambda_i}{a+\lambda_i}\right)^y$$

$$\lambda_i = \exp(x_i'\boldsymbol{\beta}) \qquad \pi_i = \frac{\exp(z_i'\boldsymbol{\gamma})}{1+\exp(z_i'\boldsymbol{\gamma})}$$

$$\Pr(y_i) = \begin{cases} \pi_i + (1-\pi_i)\left(\frac{a}{a+\lambda_i}\right)^a & y_i = 0 \\ (1-\pi_i)\frac{\Gamma(a+y)}{\Gamma(y+1)\Gamma(a)}\left(\frac{a}{a+\lambda_i}\right)^a \left(\frac{\lambda_i}{a+\lambda_i}\right)^y & y_i = 1, 2, \ldots \end{cases}$$

with:

$$E(y_i) = \mu_i = (1-\pi_i)\lambda_i$$

$$\mathrm{Var}(y_i) = \mu_i + \left(\frac{\pi_i}{1-\pi_i} + \frac{1/a}{1-\pi_i}\right)\mu_i^2 = (1-\pi_i)\lambda_i + (\pi_i + 1/a)(1-\pi_i)\lambda_i^2$$

Zero inflated models

ZIP and ZIBN regression models: Example

```
/* ESTIMATING THE ZIP MODEL */
proc genmod data=lib.simsin;
model claim=AUTO AGE_1 AGE_2 AGE_3 / dist=zip offset=lnexpos;
zeromodel  NCD_1 NCD_2 ;
run;

proc countreg data=lib.simsin;
model claim=AUTO AGE_1 AGE_2 AGE_3 / dist=zip method=qn offset=lnexpos;
zeromodel claim ~ NCD_1 NCD_2;
run;

/* ESTIMATING THE ZIBN MODEL */
proc countreg data=lib.simsin;
model claim=AUTO AGE_1 AGE_2 AGE_3 / dist=zinb method=qn offset=lnexpos;
zeromodel claim ~ NCD_1 NCD_2 ;
run;
```

Zero inflated models

ZIP and ZIBN models): Example

Procedimiento GENMOD

Información del modelo

Conjunto de datos	LIB.SIMSIN
Distribución	Zero Inflated Poisson
Función de vínculo	Log
Variable dependiente	claim
Variable Offset	lnexpos

Número de observaciones leídas	10000
Número de observaciones usadas	10000

Criterios para valorar la bondad de ajuste

Criterio	DF	Valor	Valor/DF
Desviación		20051.7637	
Desviación escalada		20051.7637	
Chi-cuadrado de Pearson	9992	13715.9629	1.3727
Pearson X2 escalado	9992	13715.9629	1.3727
Verosimilitud log		-3565.7361	
Verosimilitud log completa		-10025.8818	
AIC (mejor más pequeño)		20067.7637	
AICC (mejor más pequeño)		20067.7781	
BIC (mejor más pequeño)		20125.4464	

Algoritmo convergido.

Zero inflated models

ZIP and ZIBN models: Example

Análisis de estimadores de parámetros de máxima verosimilitud

Parámetro	DF	Estimador	Error estándar	Wald 95% Límites de confianza		Chi-cuadrado de Wald	Pr > ChiSq
Intercept	1	0.7259	0.0783	0.5723	0.8794	85.86	<.0001
AUTO	1	0.3046	0.0260	0.2537	0.3556	137.40	<.0001
AGE_1	1	0.8763	0.0860	0.7077	1.0449	103.81	<.0001
AGE_2	1	0.6807	0.0784	0.5270	0.8344	75.33	<.0001
AGE_3	1	0.3828	0.0793	0.2274	0.5382	23.31	<.0001
Escal	0	1.0000	0.0000	1.0000	1.0000		

NOTE: El parámetro de escala se ha mantenido fijo.

Análisis de estimadores de parámetros de inflación cero de máxima verosimilitud

Parámetro	DF	Estimador	Error estándar	Wald 95% Límites de confianza		Chi-cuadrado de Wald	Pr > ChiSq
Intercept	1	-0.1487	0.0362	-0.2195	-0.0778	16.90	<.0001
NCD_1	1	2.8104	0.1002	2.6140	3.0068	786.70	<.0001
NCD_2	1	1.2899	0.0666	1.1593	1.4204	375.16	<.0001

Zero inflated models

ZIP and ZIBN models: Example

Procedimiento COUNTREG

Sumarización de ajuste del modelo

Variable dependiente	claim
Número de observaciones	10000
Conjunto de datos	LIB.SIMSIN
Modelo	ZIP
Variable Offset	lnexpos
Función de vínculo ZI	Logistic
Verosimilitud log	-10026
Gradiente absoluto máximo	6.65678E-6
Número de iteraciones	19
Método de optimización	Cuasi-Newton
AIC	20068
SBC	20125

Algoritmo convergido.

Estimadores de parámetros

Parámetro	DF	Estimador	Error estándar	Valor t	Aprox Pr > |t|
Intercept	1	0.725869	0.078334	9.27	<.0001
AUTO	1	0.304624	0.025987	11.72	<.0001
AGE_1	1	0.876320	0.086010	10.19	<.0001
AGE_2	1	0.680661	0.078423	8.68	<.0001
AGE_3	1	0.382783	0.079278	4.83	<.0001
Inf_Intercept	1	-0.148661	0.036163	-4.11	<.0001
Inf_NCD_1	1	2.810377	0.100199	28.05	<.0001
Inf_NCD_2	1	1.289856	0.066593	19.37	<.0001

Zero inflated models

ZIP and ZIBN models: Example

Procedimiento COUNTREG

Sumarización de ajuste del modelo

Variable dependiente	claim
Número de observaciones	10000
Conjunto de datos	LIB.SIMSIN
Modelo	ZINB
Variable Offset	lnexpos
Función de vínculo ZI	Logistic
Verosimilitud log	-9278
Gradiente absoluto máximo	0.0002365
Número de iteraciones	31
Método de optimización	Cuasi-Newton
AIC	18574
SBC	18639

Algoritmo convergido.

Estimadores de parámetros

Parámetro	DF	Estimador	Error estándar	Valor t	Aprox Pr > |t|
Intercept	1	0.217220	0.114398	1.90	0.0576
AUTO	1	0.425502	0.043182	9.85	<.0001
AGE_1	1	1.068571	0.132415	8.07	<.0001
AGE_2	1	0.744357	0.108133	6.88	<.0001
AGE_3	1	0.361128	0.108705	3.32	0.0009
Inf_Intercept	1	-1.519991	0.241383	-6.30	<.0001
Inf_NCD_1	1	3.715311	0.222263	16.72	<.0001
Inf_NCD_2	1	2.045987	0.194797	10.50	<.0001
_Alpha	1	1.234649	0.128477	9.61	<.0001

Zero inflated models

ZIP and ZIBN models: Example

By using R:

```
># we install "PSCL"
>install.packages("pscl")
>library(pscl)

>#  ESTIMATING THE ZIP MODEL;
>regZIP <- zeroinfl(claim ~ AUTO+AGE_1+AGE_2+AGE_3 | NCD_1+NCD_2,offset=lnexpos,
+                    data=Simsin, dist="poisson", link="logit")
summary(regZIP)
>#  ESTIMATING THE ZIBN MODEL;
>regZIBN <- zeroinfl(claim ~ AUTO+AGE_1+AGE_2+AGE_3 | NCD_1+NCD_2,offset=lnexpos,
+                    data=Simsin, dist="negbin", link="logit")
summary(regZIBN)
```

Zero Inflated models

ZIP and ZIBN models: Example

```
Call:
zeroinfl(formula = claim ~ AUTO + AGE_1 + AGE_2 + AGE_3 | NCD_1 + NCD_2,
data = Simsin, offset = lnexpos, dist = "poisson", link = "logit")

Pearson residuals:
     Min       1Q   Median       3Q      Max
-0.93672 -0.49756 -0.24035 -0.12633 20.37478

Count model coefficients (poisson with log link):
            Estimate Std. Error z value  Pr(>|z|)
(Intercept) 0.725868   0.078334  9.2663 < 2.2e-16 ***
AUTO        0.304625   0.025988 11.7220 < 2.2e-16 ***
AGE_1       0.876322   0.086010 10.1886 < 2.2e-16 ***
AGE_2       0.680662   0.078423  8.6794 < 2.2e-16 ***
AGE_3       0.382786   0.079278  4.8284 1.376e-06 ***

Zero-inflation model coefficients (binomial with logit link):
             Estimate Std. Error z value  Pr(>|z|)
(Intercept) -0.148658   0.036163 -4.1108 3.943e-05 ***
NCD_1        2.810376   0.100198 28.0482 < 2.2e-16 ***
NCD_2        1.289854   0.066593 19.3691 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Number of iterations in BFGS optimization: 16
Log-likelihood: -10026 on 8 Df
```

Zero inflated models

ZIP and ZIBN models: Example

```
Call:
zeroinfl(formula = claim ~ AUTO + AGE_1 + AGE_2 + AGE_3 | NCD_1 + NCD_2,
data = Simsin, offset = lnexpos, dist = "negbin", link = "logit")

Pearson residuals:
     Min       1Q   Median       3Q      Max
-0.71712 -0.42564 -0.20785 -0.11899 19.01833

Count model coefficients (negbin with log link):
             Estimate Std. Error z value  Pr(>|z|)
(Intercept)  0.217222   0.114399  1.8988 0.0575889 .
AUTO         0.425502   0.043182  9.8538 < 2.2e-16 ***
AGE_1        1.068569   0.132415  8.0698 7.039e-16 ***
AGE_2        0.744355   0.108133  6.8837 5.831e-12 ***
AGE_3        0.361126   0.108705  3.3221 0.0008935 ***
Log(theta)  -0.210787   0.104061 -2.0256 0.0428057 *

Zero-inflation model coefficients (binomial with logit link):
            Estimate Std. Error z value  Pr(>|z|)
(Intercept) -1.51999    0.24139 -6.2968 3.038e-10 ***
NCD_1        3.71531    0.22227 16.7154 < 2.2e-16 ***
NCD_2        2.04599    0.19480 10.5029 < 2.2e-16 ***
---
Signif. codes:  0 '***' 0.001 '**' 0.01 '*' 0.05 '.' 0.1 ' ' 1

Theta = 0.80995
Number of iterations in BFGS optimization: 31
Log-likelihood:  -9278 on 9 Df
```

Data analytics for non life insurance (I)
GLM

Material created by PhD. Ramon Alemany

Department of Econometrics, Statistics and Applied Economics

MESIO